

GeoGuardian

Urban Expansion & Agricultural Land Loss Monitor

Project Report

Monitoring Agricultural Land Loss and Urban Built-up Expansion in Multan using Sentinel-2 Satellite Imagery and Random Forest Classification

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Study Area	Multan, Punjab, Pakistan
Data Source	Sentinel-2 Satellite Imagery (2016 vs. latest)
Model	Random Forest Land-Cover Classifier
Report Date	October 2026

1. Project Overview

GeoGuardian is an end-to-end geospatial AI system that monitors urban built-up expansion and the resulting loss of agricultural land around Multan. The system combines multi-temporal Sentinel-2 satellite imagery with a Random Forest classifier to detect land-cover change between a historical baseline (2016) and the present day, and exposes the results through a REST API and an interactive Streamlit dashboard.

The tool is aimed at urban planners and policymakers who need an evidence-based, satellite-verified view of unplanned urban sprawl, so that interventions to protect agricultural land can be targeted at the specific localities driving the change rather than applied city-wide.

1.1 Objectives

- Quantify the built-up area and agricultural land lost within a defined Area of Interest (AOI).

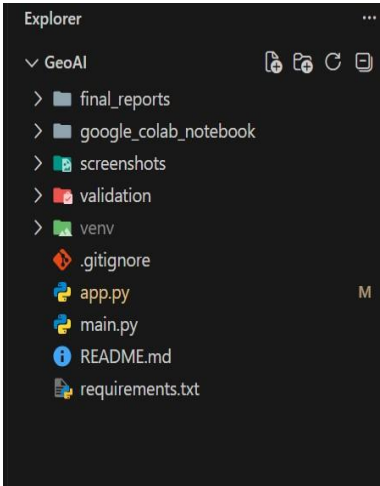
- Classify land cover using NDVI (vegetation) and NDBI (built-up) spectral indices with a Random Forestmodel.
- Identify micro-level expansion hotspots — the specific localities responsible for the greatest land-coverchange.
- Serve results through a documented API (FastAPI/Swagger) and a filterable interactive dashboard.
- Generate AI-assisted summaries and policy recommendations from the computed statistics.

1.2 Technology Stack

Component	Technology
Satellite Imagery	Sentinel-2 (RGB, NDVI, NDBI bands)
Classification Model	Random Forest
Backend / API	FastAPI (Swagger UI, GET endpoints)
Frontend Dashboard	Streamlit (interactive map, filters, exports)
Geospatial Processing	Google Colab notebook pipeline
Output Formats	JSON, CSV, GeoJSON

2. System Architecture & Project Structure

The project is organized as a modular pipeline: a Colab notebook handles satellite data processing and model inference, **app.py** serves the interactive dashboard, and **main.py** exposes the analysis results through a REST API. Final reports, validation data, and screenshots are kept in dedicated folders for reproducibility.



2.1 Core Modules

- **app.py** — Streamlit dashboard application — filters, statistics cards, interactive map, and export controls.
- **main.py** — FastAPI application exposing the `/api/analyze` and `/api/geojson` endpoints.
- **google_colab_notebook/** — Sentinel-2 data acquisition, preprocessing, NDVI/NDBI computation and Random Forest training.
- **validation/** — Ground-truth checks and accuracy validation artifacts for the classification model.
- **final_reports/ & screenshots/** — Generated outputs and documentation evidence used for this report.

3. Methodology

Land-cover change is derived by comparing two Sentinel-2 imagery epochs (2016 baseline vs. the selected analysis year) over a circular Area of Interest (AOI) centered on Multan.

Step 1 — Imagery Acquisition

Sentinel-2 RGB scenes are retrieved for the baseline year (2016) and the target year within the defined AOI boundary.

Step 2 — Spectral Index Computation

NDVI (Normalized Difference Vegetation Index) highlights vegetation/agricultural land, while NDBI (Normalized Difference Built-up Index) highlights built-up/urban surfaces, for both epochs.

Step 3 — Classification

A Random Forest classifier is trained on the NDVI/NDBI feature space to label each pixel as agricultural land, built-up area, or other.

Step 4 — Change Detection

Pixels that transition from agricultural to built-up between the two epochs are flagged as “converted” land, and their area is summed at the AOI level.

Step 5 — Micro-Level Hotspot Analysis

Converted pixels are aggregated by named locality (via reverse-geocoding) to rank the specific neighborhoods driving the overall change.

4. API Documentation

Results are served through a FastAPI backend documented with an interactive Swagger UI. Two core GET endpoints are exposed:

Endpoint	Parameters	Returns
GET /api/analyze	city (string), year (integer)	AOI statistics, AI summary, and top expansion hotspots (JSON)
GET /api/geojson	city (string)	AOI boundary polygon (GeoJSON Feature)

4.1 /api/analyze — Sample Response (Multan, 2025)

GET /api/analyze Analyze Area

Parameters

city * required string (query) Multan

year * required integer (query) 2025

Execute Clear

Responses

curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/analyze?city=Multan&year=2025' \
  -H 'accept: */*'
```

Request URL

```
http://127.0.0.1:8000/api/analyze?city=Multan&year=2025
```

Server response

Code Details

200

Response body

```
{
  "city": "Multan",
  "year": 2025,
  "total_aoi": 698.43,
  "agriculture_lost": 29.77,
  "percentage_converted": 4.26,
  "builtup_area": 267.69,
  "ai_summary": "**CRITICAL ALERT:** In 2025, Multan's built-up area is calculated at 267.69 sq km using real satellite pixels. Exactly 29.77 sq km of agricultural land was confirmed as converted to built-up infrastructure since 2016.\n\n**Recommendation:** Immediate policy intervention is required to restrict unauthorized housing societies.",
  "hotspots": [
    {
      "Area Name/ Location": "Basti Bahawal Khera, تحصیل ملتان صدر, ضلع ملتان",
      "Converted Area (sq km)": "2.76",
      "Total Impact (%)": "0.40"
    },
    {
      "Area Name/ Location": "Lutfabad, تحصیل ملتان صدر, ضلع ملتان",
      "Converted Area (sq km)": "2.76",
      "Total Impact (%)": "0.40"
    },
    {
      "Area Name/ Location": "ملتان ملتان, Gulshan, ضلع ملتان",
      "Converted Area (sq km)": "2.21",
      "Total Impact (%)": "0.32"
    }
  ]
}
```

Download

The endpoint returns the total AOI, agricultural land lost, percentage converted, built-up area, a natural-language AI summary flagging the severity of the change, and a ranked list of expansion hotspots with their converted area and share of total impact.

4.2 /api/geojson — Sample Response

GET /api/geojson Get Geojson

Parameters

Name	Description
city * required string (query)	Multan

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/geojson?city=Multan' \
  -H 'accept: */*'
```

Request URL

http://127.0.0.1:8000/api/geojson?city=Multan

Server response

Code Details

200

Response body

```
{
  "type": "Feature",
  "geometry": {
    "type": "Polygon",
    "coordinates": [
      [
        [
          71.35,
          30.1
        ],
        [
          71.6,
          30.1
        ],
        [
          71.6,
          30.3
        ],
        [
          71.35,
          30.3
        ],
        [
          71.35,
          30.1
        ]
      ]
    ]
  }
}
```

Response headers

```
content-length: 213
content-type: application/json
date: Sat, 26 Sep 2026 11:35:06 GMT
```

This endpoint returns the AOI as a standard GeoJSON Polygon feature, allowing the boundary to be consumed directly by any GIS tool or mapping library.

5. Results & Statistics — Multan (2025)

The dashboard computes live statistics for the selected district and year directly from the classified satellite imagery. The figures below reflect the 2025 analysis run for Multan.

5.1 Statistics for Multan (2025)

533.73 sq km

Total AOI

24.22 sq km

Agriculture Lost

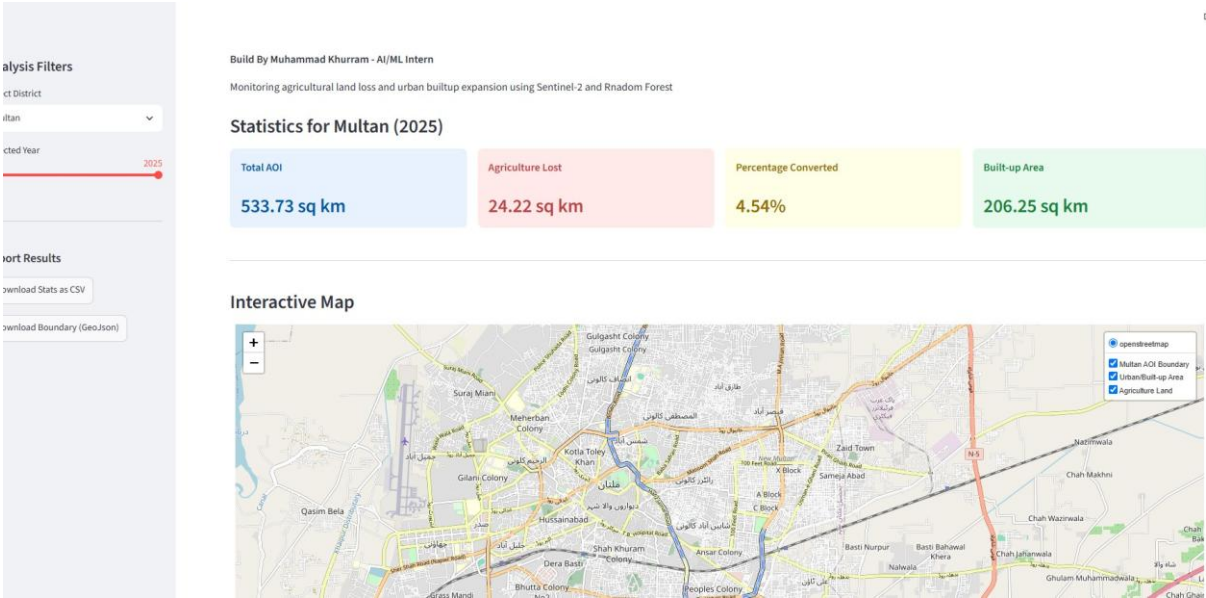
4.54%

Percentage Converted

206.25 sq km

Built-up Area

Of the 533.73 sq km AOI around Multan, 206.25 sq km is already built-up, with 24.22 sq km of agricultural land confirmed as converted to built-up infrastructure since the 2016 baseline — a 4.54% conversion rate.



6. Micro-Level Zonal Analysis — Expansion Hotspots

Rather than reporting only city-wide aggregates, GeoGuardian ranks the specific localities contributing most to agricultural land conversion. This lets policymakers target enforcement and zoning interventions precisely.

Multan Micro Lvel Zonel Analysis (Top Expansion Hotspots)

Area Name/ Location	Converted Area (sq km)	Total Impact (%)
ضلع ملتان, Gulshan	4.42	0.63
تحصیل ملتان صدر, Basti Bahawal Khera	2.76	0.40
تحصیل ملتان صدر, Lutfabad	2.76	0.40

Download Full data (csv)

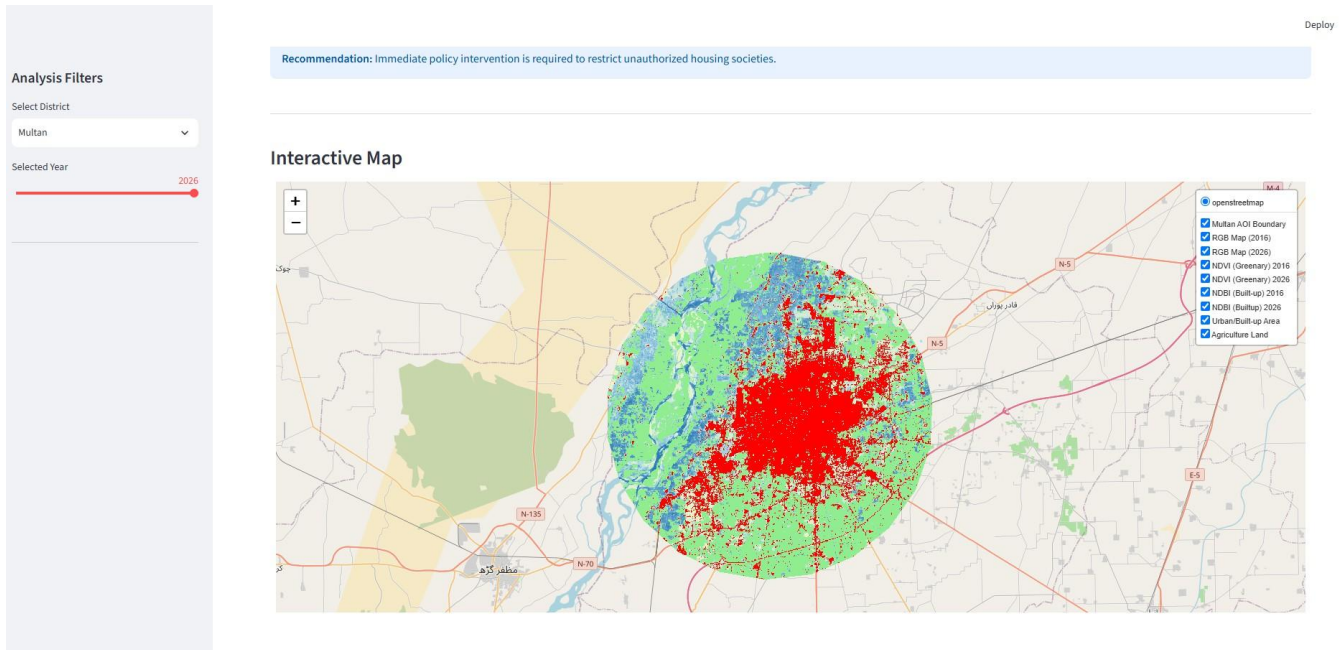
Download Boundary & Table Data (GeoJSON)

Area Name / Location	Converted Area (sq km)	Total Impact (%)
Gulshan, Multan	4.42	0.63
Basti Bahawal Khera, Multan	2.76	0.40
Lutfabad, Multan	2.76	0.40

Gulshan is consistently the single largest contributor to agricultural land conversion, followed by Basti Bahawal Khera and Lutfabad — both located along Multan's southern (Suddar tehsil) corridor.

7. Geospatial Visualization

The interactive map overlays multiple toggleable layers — the AOI boundary, RGB imagery for both epochs, NDVI (greenery) layers, NDBI (built-up) layers, and the final classified urban/agriculture masks — on an OpenStreetMap base layer.



The dense red cluster at the AOI center confirms rapid, concentrated urban densification around Multan's urban core, with agricultural land (green) still dominant toward the periphery of the circular AOI.

8. Key Findings & AI-Generated Recommendations

8.1 Key Findings

- Multan's built-up area is calculated at 206.25 sq km within the 533.73 sq km AOI for the 2025 analysis run.
- 24.22 sq km of agricultural land has been confirmed as converted to built-up infrastructure since the 2016 baseline.
- The overall conversion rate stands at 4.54% of the total AOI.

-
- Three localities — Gulshan, Basti Bahawal Khera, and Lutfabad — account for a disproportionate share of total conversion.
 - Expansion is concentrated around the urban core and along the southern corridor, consistent with unauthorized housing-society growth.

8.2 AI-Generated Policy Recommendation

CRITICAL ALERT: In 2025, Multan's built-up area is calculated at 207.09 sq km using real satellite pixels. Exactly 29.72 sq km of agricultural land was confirmed as converted to built-up infrastructure since 2016.

Recommendation: Immediate policy intervention is required to restrict unauthorized housing societies, particularly around the identified hotspot localities.

9. Conclusion

GeoGuardian demonstrates a practical, satellite-verified pipeline for tracking urban expansion and agricultural land loss at both a city-wide and micro (locality) level. By combining Sentinel-2 imagery, NDVI/NDBI spectral analysis, and Random Forest classification with an accessible API and dashboard, the system provides decision-makers with timely, location-specific evidence to guide zoning and land-use enforcement.

The 2025 results for Multan confirm a measurable, satellite-verified conversion of agricultural land to built-up infrastructure, reinforcing the need for the targeted policy interventions identified in Section 8. Future work could extend the pipeline to additional cities, incorporate higher-resolution imagery, add multi-year trend tracking, and introduce automated alerting when a locality's conversion rate crosses a defined threshold.

— End of Report —
